Python Assignment
Topic-9: String Manipulation

Q1. Understanding how to access and manipulate strings.

Ans.

Strings are used to store text in Python. We can access characters in a string using indexing. The first character starts from index 0. We can also manipulate strings using different operations like changing case, joining strings, or replacing characters. Strings are immutable, which means they cannot be changed directly. Instead, new strings are created. String manipulation is very useful in programs where text processing is required. It is commonly used in user input and data handling.

Example:

```
name = "Python"
print(name[0])
print(name.lower())
```

Q2. Basic operations: concatenation, repetition, string methods.

Ans.

Python provides many basic operations for working with strings. Concatenation is used to join two strings using the + operator. Repetition is done using the * operator to repeat a string multiple times. String methods like upper(), lower(), replace(), and strip() help in modifying strings. These operations make string handling easy and flexible. They are commonly used in text processing and formatting.

Example:

```
a = "Hello"
b = "World"
print(a + b)
print(a * 2)
print(a.upper())
```

Q3. String slicing.

Ans.

String slicing is used to extract a part of a string. It is done using index values inside square brackets. We can specify starting index and ending index to get a substring. Slicing helps in accessing a range of characters easily. It is useful in many tasks like extracting data or formatting strings. If no index is given, Python takes default values. Slicing makes string handling more powerful and flexible.

Example:

```
text = "Python"
print(text[1:4])
print(text[:3])
print(text[-1])
```